

Artificial intelligence and patient care: Perspectives of audiologists and speech-language pathologists

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ABSTRACT

Background: Artificial Intelligence has been implemented across various fields, including healthcare, where it has significantly advanced patient care in recent years. The present study aimed to explore the perspectives of audiologists and speech-language pathologists (ASLPs) toward AI in patient care.

Methods: The study employed a cross-sectional design with a convenience sampling method. The questionnaire included 27 questions consisting of demographic details and perspectives towards AI in audiology and speech language pathology services. Descriptive statistics were performed to analyze the data.

Results: Ninety-five ASLPs participated in the study, working across different work settings and with a mean age of 28.34 years, ranging between 18 and 47 years. Almost 50 % of participants reported AI tools can be helpful in diagnosis and planning the treatment. About One-fourth (25 %) believed that AI could help in rehabilitation. Few of participants (14.8 %) reported that AI may replace audiology and speech-language pathology services. ChatGPT was the most used platform by ASLPs in their practice. The ASLP clinicians believed AI would revolutionise ASLP practice without alarming effects on their employability.

Conclusion: The findings suggest that while AI has potential in ASLP practice, there is still a need for greater understanding and adoption of the technology.

1. Introduction

Artificial Intelligence (AI) is an innovation of machine and deep learning that encompasses a comprehensive simulation of human intelligence and behaviors. Machine learning facilitates the system to perform fundamental data analysis, and deep learning uses intricate structures to create models [1]. The simulation of human behavior includes acquiring information, developing rules for securing the data, eliciting approximate or definite conclusions and self-correction [2]. The AI functions include automated interfaces for speech recognition, visual perception, decision-making, and language translation [3]. The rapid expansion of AI has been implemented in several fields such as agriculture, cybersecurity, education, entertainment finance, games, healthcare, industries, and many more [4]. This evolving technology has started growing in healthcare as well.

The origin of AI medical applications began in the 1950s to mimic human decisions and actions. Since then, several revolutionary

innovations have developed such as the first robotic arm in 1955 by General Motors and the first chatterbot by Joseph Weizenbaum in 1964 at the MIT AI Laboratory that could emulate conversation with a human therapist [5]. In the 1980s, another advancement was designed at the University of Massachusetts, a program named DXplain to support clinicians in landing a medical diagnosis. In recent years, AI tools have been developed to analyze a wide range of health data comprising patient data from multi-omic techniques, clinical, behavioral, environmental, and drug data [1]. Medical imaging to store and collect the scans, analysis of echocardiography scans, and detection of speech samples to identify neurological conditions like Parkinson's disease are a few possible areas of AI in clinical care. Apart from these, AI applications include diagnosis and treatment suggestions, improved patient engagement and compliance and administrative procedures [6,7].

The applications of AI have been investigated in various health sectors to provide clinical services, for instance, in the fields of dentistry, ophthalmology, pharmacy and radiology [8–11]. The implementation of

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AI can also transform the field of audiology and speech-language pathology. The precision of computational audiology can benefit clinicians in diagnostic procedures, rehabilitation recommendations and comprehension of auditory systems and behaviors [12]. In the field of speech-language pathology, automatic speech recognition (an AI-driven technology) has proven beneficial in the telerehabilitation of patients with dysarthria under the remote supervision of speech-language pathologists (SLP) [13]. Likewise, recent reviews suggested that AI tools can be integrated into aphasia treatment and provide patient-centered and customized rehabilitation plans [14,15]. However, ethical considerations, violation of data privacy, liability, and scientific limitations are concerning challenges in delivering AI-driven hearing healthcare and speech-language rehabilitation services [12,14].

Several recent studies investigated the knowledge, attitude and perception towards AI in healthcare of dental students, medical students, pediatricians, and radiologists [16–19]. It is possible that audiologists and speech-language pathologists (ASLPs) may have valuable insights to implement AI successfully in clinical settings. Therefore, incorporating audiologists and speech-language pathologists could provide a better understanding to develop AI algorithms for delivering hearing healthcare services and speech-language rehabilitation. Before the commencement of AI implementation, it is significant to understand the perspectives of ASLPs toward AI in patient care. To date, no study has been reported among ASLPs that explore their perspectives on artificial intelligence (AI). This study aimed to understand the perspectives of AI in patient care among ASLPs.

2. Method

A cross-sectional study design was used, and responses were collected without employing any invasive procedure. The present study followed the Helsinki Declaration guidelines [20], the study was approved by ethical committee of Amity University, India. Audiologists and speech-language pathologists were invited to fill out an online questionnaire to explore their perspective of AI in patient care using convenience sampling.

2.1. Material

The study was conducted in two phases. A 27-item questionnaire was developed based on a review of published literature on AI [21,22]. The content validity of the developed questionnaire was assessed by five audiologists and speech-language pathologists with over five years of experience in the field. Professionals were asked to rate each item in the questionnaire for its relevance for inclusion using a four-point rating scale; not relevant, somewhat relevant, quite relevant and relevant [23]. All items were rated as quite relevant and relevant and were retained. Scale- Content validity index (S-CVI) was used to establish the content validity [24]. The S-CVI was calculated using the formula: Total score for each item per respondent/total number of items and a score of 0.86 was obtained indicative of excellent content validity. In this way, the final questionnaire was developed and made available for the next phase of the study.

The finalized questionnaire included 27 questions consisting of closed questions and Likert scale questions on demographic details, perspectives of ASLPs towards AI in the delivery of ASLP services and attitudes towards AI. These were followed by an open-ended question about preferred platforms used. The questionnaire was made available via Google Forms from April to May 2024. A brief introduction to the study's objectives was included in the Google form. An electronic informed consent form was embedded in the questionnaire; participants were required to complete it before participating in the study. The Google form link was generated and disseminated to all practicing ASLPs in India through emails and WhatsApp groups to ensure a pan-India survey.

2.2. Data analysis

Initially, raw data were obtained in an Excel spreadsheet (Microsoft Inc., Redmond, WA, USA) and then transferred to statistical program SPSS v25 (IBM Corp., Armonk, NY, USA). Descriptive analysis was performed and presented as the mean, standard deviation, percentage, and frequency of the collected data. For open-ended questions, common replies were computed.

3. Results

One hundred eleven participants responded to the survey, of whom 95 consented and completed the survey. The mean age of the participants was 28.34 years, range between 18 and 47 years. The majority of participants were in private setup (n = 79, 83.2 %), and the remaining (n = 16, 16.8 %) were in government. The demographic characteristics of ASLPs participated in the study are presented in Table 1.

A large proportion, 62 % of ASLP clinicians agreed that AI will benefit human health, while 53 % of clinicians were affirmative with the statement 'positive outcomes of AI on an individual's communication and hearing needs.' Clinicians were asked if 'AI will improve ASLP services, more particularly where experts are unavailable?' Only 44.2 % agreed, 30.5 % were unsure and 25.3 % were disagreed. More than half, 60 % clinicians were in favour of implementing AI tools in their practice. Ethical consideration and privacy issues are core principles for delivering health care services. To obtain the perception regarding it, the related question was asked 'use of AI in providing ASLP services is unethical?' Twenty-two per cent of clinicians agreed, 27.4 % were undecided, and half of the participants (50.5 %) disapproved. In relation to patient privacy, 48.4 % agreed that the utilization of AI tools can increase the risk to patient privacy, 33.7 % were unsure, and 17.9 % failed to agree. Concerning the more ASLP practice specific questions, there was a broad spectrum of agreement, as illustrated in Fig. 1.

A mixed opinion was obtained on whether the exponential growth of AI will risk jobs as an ASLP; 40 % disagreed that AI could risk their jobs, 30.5 % agreed, and 29.5 % were unsure. Only 20 % of clinicians believed that AI would not expand in ASLP services, nearly half 48.4 % disagreed, and 31.6 % were uncertain. Fifty-nine per cent of clinicians felt that a day may come when patients may approach AI instead of ASLP for services, 21 % agreed, and 20 % were unsure. The last question was

Table 1
Demographic details of participants.

		Number (Percentage)
Gender	Female	66 (69.5 %)
	Male	29 (30.5 %)
Qualification	Bachelor of audiology and speech-language pathology	50 (52.6 %)
	Masters in Audiology & speech language pathology	17 (17.9 %)
	Master of Science in Audiology	19 (20 %)
	Master of Science in Speech-language pathology	7 (7.4 %)
	PhD	2 (2.1 %)
Specialization	Audiology and Speech-Language pathology	42 (44.2 %)
	Audiology	27 (28.4 %)
	Speech-Language pathology	26 (27.4 %)
Experience (in years)	< 5 years	63 (66.3 %)
	6-10 years	10 (10.5 %)
	11-15 years	13 (13.7 %)
	16-20 years	5 (5.3 %)
	>20 years	4 (4.2 %)
Workplace	Clinic	50 (52.6 %)
	Hearing device company	3 (3.2 %)
	Hospital	34 (35.8 %)
	Rehabilitation centre	2 (2.1 %)
	School	6 (6.3 %)

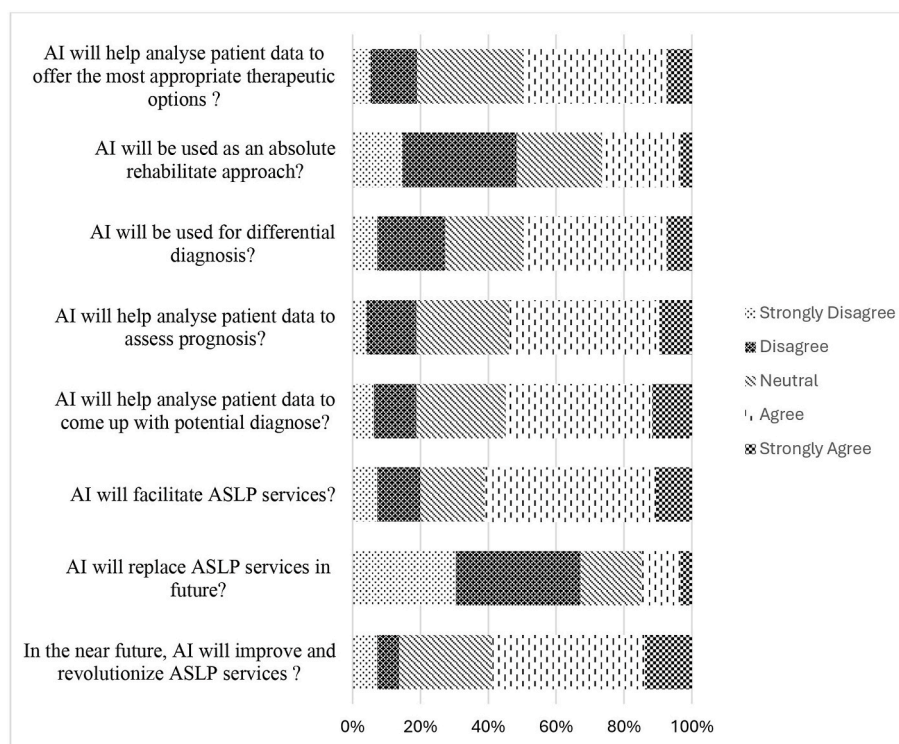


Fig. 1. Division of the percentage of agreement of the clinicians surveyed with the statements about possible applications of AI in ASLP services.

open-ended to identify AI platforms used by ASLP clinicians. The most preferred platform used was ChatGPT. Only a few participants ($n = 3$) reported another AI application called Avaz AAC.

4. Discussion

The rapid uptake of AI may lead to changes in the health sector by improving the efficiency of all kinds of procedures, from administration work to patient care. In the 2000s, the modern era of AI began with a newly discovered transformation of its application in the healthcare sector; recently, it has emerged as a powerful tool in the healthcare industry. The strength of AI in the ASLP practice lies in applications that have exhibited clinical utility in assessment, diagnosis and rehabilitation. A recent study suggested that FP-Growth and naïve Bayes algorithms (a machine learning technology) can efficiently identify hearing-loss symptoms with lesser error rates [25]. Likewise, AI can be utilized in hearing aid technology to improve speech understanding in background noise [26].

The present study was carried out to understand the perspectives of audiologists and speech-language pathologists in patient care using AI in India, with a particular interest in identifying the practical applications of AI in ASLP practice. The demographic characteristics of the ASLPs had a predominantly young population and had a bachelor's degree. Besides this, 69.5 % of the ASLPs were female, which shows the dominance of the female gender in ASLP practice. A large proportion of the ASLPs agreed that AI could benefit human health. This finding is comparable with a previous study where nursing professionals recognized the possible benefits of AI for human health and well-being [27]. Fifty-three per cent of surveyed ASLPs admitted that AI implementation can improve the communication abilities of needy individuals. Similarly, AI integration into the health sector can boost patient care in different healthcare settings [28]. This draws attention to the emergent popularity of the application of AI in the health sector.

Most ASLPs were convinced that AI would improve, revolutionize, and even facilitate ASLP services. Studies in other health sectors, like nursing, paediatrics, pharmacy and radiology, agree on this point, as

respondents believe that AI would be advantageous in their practising field [21,22,27,29]. Regarding the foreseeable future of AI application in assessment, diagnosis, and therapeutics, nearly 50 % of ASLPs accepted that AI would support analysing patient medical details to assess diagnosis. In contrast, only 25 % of ASLPs agreed that AI would be utilized as an absolute rehabilitation method. Recent studies conducted on paediatricians and radiologists showed almost similar findings related to diagnosis, appropriate therapeutic options, detection of disease via images, and indication of corresponding radiological examinations [21,22]. These findings indicate possibilities of acceptance and adoption of AI tools in ASLP practice. Only a few (14.8 %) ASLPs responded optimistically on whether AI will replace ASLP services in future. This is closely linked to the findings of the previous studies, where respondents considered AI a non-viable option for delivering health services [22,30]. Consequently, AI is developed to complement and enhance skills to improve the quality of healthcare services rather than replace health professionals [31].

This study reveals a mixed response among ASLPs on whether AI could risk their jobs as an ASLP. AI may enable the automation of tailored treatment plans and audiological procedures like an otoscopic diagnosis of outer and middle ear disorders and automatic speech recognition; however, it is unlikely that AI can jeopardize the career of an ASLP [32]. Moreover, issues like job security should be addressed through open communication among ASLPs, and AI technology experts. Subsequently, 48.5 % of ASLPs were affirmative that AI would be prevalent in ASLP services. These findings are consistent with an earlier study investigating which pharmacist's responses and respondents agreed pharmaceutical AI may rise in the future [29]. Perhaps, AI can be incorporated into healthcare services, but only under the supervision of health professionals [31]. Data privacy and ethical concerns are major drawbacks associated with using AI in any field, including healthcare. Almost 48 % of ASLPs in this study stated that using AI might compromise patient data. A recent study surveying paediatricians in France showed similar findings, where respondents agreed AI may threaten patient data security [21]. To deal with these kinds of challenges, appropriate safeguards are essential to maintain autonomy and

confidentiality and prevent data breaches that may occur from AI-driven algorithms [33]. Ethical practice promotes accountability for the offered treatment and commitment towards patients and caregivers by health professionals. Concerning ethical considerations, only 22 % of ASLPs agreed that AI inclusion was unethical in ASLP practice. Besides this, paediatricians in France perceived that AI was harmless to ethical and human dimensions of medicine [29]. Undoubtedly, health-related AI tools should be developed to ensure patient safety and privacy. Fifty-nine per cent of ASLPs disagreed that individuals might prefer AI over ASLPs. Over the next few years, AI may assist in the fitting of hearing aids or the planning of therapies, whereas counselling, decision-making based on the varied spectrum of patients, and a personalized approach are beyond the reach of AI [32].

The most commonly used AI chatbot tool was ChatGPT. Almost every ASLP was using this platform based on his/her needs. ChatGPT has emerged as one of the more established AI tools across the world, since it aids in numerous tasks, including text generation, translation, and data analysis [34]. Nevertheless, ChatGPT has limitations, for instance, biased or inappropriate content, data reliability, and misleading information. Furthermore, ChatGPT can be adopted in the health sector to generate automated summaries by dictation, receive updates, and rule out potential diagnoses based on signs and symptoms [35]. Avaz AAC was another application reported by only three ASLPs. Avaz AAC enables individuals to communicate their needs, emotions, and ideas. Unexpectedly, only a few ASLPs were aware of the profession-related applications, this gap could be due to the lack of awareness about available tools. As a result, the findings highlight the need to create more awareness and training on AI-related courses among ASLPs to improve knowledge and skills, which could lead to more practical application of AI in ASLP practice. Nevertheless, AI technology may significantly alter ASLP clinical services; to attain a definitive choice, AI should be developed with superior processing and conceptualization characteristics to become an integral tool of ASLP practice.

5. Limitations

This study comprises some valuable insights about the perspectives of AI in patient care among ASLPs; however, it does have its limitations. This study is specific to Indian ASLPs; it would be interesting to investigate perspectives across diverse countries with a larger sample size. The study findings also indicate the need for qualitative studies such as interviews and focus groups to obtain deeper insights into ASLPs related to AI in delivering ASLP services.

6. Conclusion

This study found that ASLPs reflected AI may revolutionize and improve ASLP services. More than half of ASLPs believed that the implementation of AI could improve the communication abilities of needed individuals. Participants responded that differential diagnosis, prognosis, and suggestions for suitable therapeutic techniques were the domains of ASLP services where AI would be most helpful. Most of the participants disagreed that AI could replace the ASLP services. Overall, findings suggest that AI can be perceived as a potential tool for ASLPs to improve the quality of patient care and advance ASLP practice. However, it is essential to continue monitoring the impact of AI on ASLP practice and job security to address any potential issues and ensure that the integration of AI into ASLP practice is done ethically. Based on these results, workshops and seminars must be organized to ensure that ASLPs gain a better comprehension of AI and ultimately facilitate them to play a fully conscious role in the development and implementation of AI tools in ASLP practice. Since AI continues to restructure the healthcare landscape, the perspectives of ASLPs will play a decisive role in shaping its impact on patient care. Besides this, future research should focus on identifying the challenges and facilitators to the incorporation of AI technology in ASLP practice.

CRedit authorship contribution statement

Komal Aggarwal: Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Data curation, Conceptualization. **Rohit Ravi:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Krishna Yerraguntla:** Writing – review & editing, Supervision, Resources, Project administration, Formal analysis, Conceptualization.

Disclosure statement

No potential conflicts of interest were reported by the author(s).

Statement of ethics

The study was approved by ethical committee of Amity University, India.

Data availability statement

All data generated or analysed during the study are included in this article. Further enquires can be directed to corresponding author.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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